

M BILAL FAISAL

Career Context Document — Source of Truth for CVs, Portfolio and Profiles

Version 2.2 · Companion to the four final CVs · August 2026

HOW TO USE THIS DOCUMENT

This is the single source of truth behind the four CVs. Every fact here has been checked and is safe to reuse in a résumé, a LinkedIn profile, a portfolio site, a cover letter or an interview answer.

If you hand this to an AI tool to build a portfolio, a bio or another CV variant, paste the whole document. The exclusions and guardrails at the end are load-bearing: without them a model will pad the material with invented metrics.

Hard rule for any tool consuming this document: use only the numbers written here. Do not generate percentages, user counts, latency figures, accuracy deltas, revenue impact or time savings. None of those have been measured.

IDENTITY AND CONTACT

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GitHub: github.com/lambilalfaisal

Portfolio: portfolio-pi-peach-78.vercel.app

Availability: Open to remote full-time roles

POSITIONING

Four CV variants exist, in order of strength:

- **AI / ML Engineer — primary.** Leads on RAG, LLM and agentic systems in production. This is the strongest framing because the production AI work is genuinely differentiated for a candidate at this stage.
- **Full-Stack Engineer.** Leads on .NET, React and SQL Server delivery, with production AI as depth rather than headline. Use for conventional product engineering roles.
- **Data Scientist.** Leads on the ML, evaluation and data-engineering side. Use where the role is analysis-first or the degree matters more than the platform work.
- **Master CV.** Complete inventory. Not for submission — it is the pool the other three are cut from.

Consistent through all four: builds intelligent systems that actually reach production, and builds the conventional software those features have to live inside. That combination is the story. Neither half alone is as strong.

VERIFIED FACTS

Employment — Acme One, Lahore, Pakistan

A technology consulting and solutions provider building Nucleus One, a multi-module enterprise platform.

Associate Software Engineer (full-time): March 2026 to present

Artificial Intelligence Intern: July 2025 to March 2026

The internship converted early into a full-time engineering offer. Total professional experience: over a year, continuous, at one company.

Education

Degree: Bachelor of Science in Data Science, 2023 to 2027, final year

Institution: University of Management and Technology (UMT), Lahore, Pakistan

Coursework: Machine learning, deep learning, artificial intelligence, information security, database systems, data structures and algorithms

Research focus: Transformer architectures, large language models, retrieval-augmented generation systems

Studied alongside full-time engineering work. Worth stating — it explains the overlap between degree dates and employment dates before an interviewer has to ask.

Certifications

- Anthropic, 2026: Claude Code in Action; Claude Code 101; AI Fluency: Framework and Foundations; AI Capabilities and Limitations; Claude 101

Removed as of this version: Microsoft AI-103. It is no longer listed on any CV, and the sentence "Studying for Microsoft AI-103" has been cut from both the AI/ML resume summary and the Master CV summary. Do not reintroduce it anywhere — including a portfolio site — unless and until the exam is actually passed.

Languages

English (professional working proficiency), Urdu (native), Punjabi (native), Arabic (reading)

Interests

Strength training and distance running, cricket, futsal, badminton, table tennis; reading across behavioural economics, psychology and technology history

THE ONLY NUMBERS THAT MAY BE USED

These are verified. Every other figure is off-limits.

- Roughly 100 tables in the unified SQL Server schema shared across Nucleus One modules
- Ranked second by commit volume across both platform repositories
- Seven-person team
- Sole engineer on one full platform module (Project-One), database through frontend
- Roughly ten custom tools across the two RepoWarden agents
- Two agents in RepoWarden, built as two compiled StateGraphs
- Two-pass verification in Project-Ease (a second, temperature-zero model call)
- Dates: internship July 2025 to March 2026; full-time March 2026 to present; degree 2023 to 2027

WORK AT ACME ONE — FULL DETAIL

Associate Software Engineer, March 2026 to present

- Builds LangChain and LangGraph agentic pipelines in Nucleus One: multi-step chains, tool-calling agents with custom tool definitions, stateful graphs, and human-in-the-loop approval steps, integrated with the platform's ASP.NET Core 8 APIs and SQL Server data layer. These pipelines run inside the Nucleus One production platform — confirmed.
- Implemented LiveKit voice agents for a real-time AI assistant on a construction-site management platform, building the agent worker architecture handling live audio streams and LLM-driven responses over WebRTC.
- Uses Azure AI Foundry for model management and evaluation, and runs Llama 3.1 locally through Ollama for offline inference and pipeline testing.
- Sole engineer on the Project-One module, delivered end to end from SQL Server schema through ASP.NET Core 8 services to the React/TypeScript interface: Epic to Feature to Story to Task hierarchies, sprint planning, Kanban boards, backlog management, analytics dashboards. Module features integrate with existing entities on the shared schema rather than standing alone.
- Authored the HR-One service layer covering onboarding, offboarding and evaluation workflows, and deployed its SQL Server schema.

- Built the platform authentication controller with JWT, RBAC and secure session handling.
- Shipped TimeTrace to production with its backend reporting engine, and tuned queries and indexes for reporting workloads.
- Implemented client-side state management with Zustand and React Query; built responsive interfaces with Tailwind CSS; expanded frontend breadth across Next.js, Vue.js and Laravel through additional client-facing project work.
- Participates in code review, sprint planning and technical design discussions; maintains technical documentation across owned modules.

Artificial Intelligence Intern, July 2025 to March 2026

- Engineered and deployed a Retrieval-Augmented Generation chatbot into Nucleus One's production environment, letting employees ask open-ended questions in natural language and get answers drawn from live platform data across the roughly 100-table SQL Server schema. Owned the feature from prototype through go-live.
- Made retrieval permission-aware: every request is resolved against the asking user's existing authorisation in the application, so the chatbot can only surface records that user is already entitled to see and cannot become a path around the platform's access controls. This is the single most interview-ready detail in the whole document — it shows security thinking, not just prompt plumbing.
- Improved retrieval grounding through prompt engineering and iteration on chunking strategy and embedding configuration, measured against answer accuracy on internal documents.
- Built and evaluated NLP and text-classification models in Python with pandas, NumPy and scikit-learn — data cleaning, exploratory analysis, feature engineering, model comparison — then deployed the selected sentiment model behind REST endpoints in a production web application.
- Performed end-to-end data preparation on internal company datasets and iterated on feature selection and model choice to improve classification accuracy.

Nucleus One modules touched

- **Project-One:** work item tracking with hierarchical Epic, Feature, Story and Task structures, sprint planning, Kanban boards, backlog management, analytics dashboards. Sole engineer.
- **HR-One:** employee profile management, department and designation hierarchies, employment history, attendance tracking, HR metrics dashboards. Authored the service layer and deployed the schema.
- **Ops-One:** asset lifecycle management with QR code tracking, automated expense processing with receipt scanning, approval workflows, reimbursement processing.
- **TimeTrace:** production time-tracking module with a backend reporting engine. Shipped to production.

PROJECTS

RepoWarden — Autonomous Code Review and Refactoring Agents

Python, LangGraph, LangChain, FastAPI, MongoDB, React 19, Vite, Monaco · github.com/lambilalfaisal/RepoWarden

Two-agent LangGraph system that reviews, evaluates and safely refactors a codebase, built as two compiled StateGraphs behind a FastAPI backend.

- Capability-based agent isolation: the Reviewer agent's toolset structurally omits the write tool, so it cannot modify the workspace regardless of how it is prompted. Safety enforced through tool binding rather than instructions.
- Human-in-the-loop authorisation gate: nothing is written to disk until the user approves, at which point the write-capable Editor agent is constructed and granted the write tool.
- Custom tool set covering directory listing, file reads, code search, dedicated security, performance and code-quality analysers, and structured edit proposals — roughly ten tools across the two agents — over a sandboxed filesystem layer.
- A pytest backend suite plus a fixture directory of deliberately flawed code for end-to-end agent evaluation.
- Token-by-token Server-Sent Events streaming, with structured events for tool calls, analysis results and proposed edits.

- MongoDB-backed persistence: a LangGraph checkpointing for thread-scoped conversation state plus a custom store for cross-session project memory, removing any need to replay history from the client.

Why it matters: the capability-isolation argument is the most sophisticated single idea in the portfolio. Enforcing safety structurally rather than through prompt instructions is a real engineering position, and it will land with senior interviewers.

Project-Ease — Multi-Tenant AI Platform for Law Firms

Python, Quart, Azure OpenAI, Azure AI Search, Azure AI Document Intelligence, RAGAS, React 19 · github.com/lambilalfaisal/Project-Ease

In development. Extension of Microsoft's azure-search-openai-demo reference architecture into a legal practice platform for Pakistani law firms. Always describe it as in development — never as shipped.

- Multi-tenant retrieval: organisation-scoped isolation alongside a shared case-law corpus, resolved in a single hybrid BM25 plus vector plus semantic query, with tenancy enforced server-side from the JWT organisation claim rather than trusting any client-supplied value.
- Two-pass anti-hallucination verification: a second temperature-zero model call checks every answer against its retrieved sources before the user sees it.
- RAGAS evaluation harness wired into the response path, scoring faithfulness, answer relevancy and context precision, so retrieval and prompt changes can be compared against measured output quality rather than impressions. (See guardrail 2 — the harness exists; the scores do not yet.)
- Practice-management and SaaS layer: matter lifecycle tracking, hearing and deadline management, OCR-assisted document extraction, statutory fee calculation, WhatsApp/Twilio reminders with Whisper voice transcription, per-seat plan enforcement, audit logging, and an Urdu/RTL interface with language-matched prompting.

Multi-Factor Authentication Server

Python, scikit-learn, Streamlit, WebAuthn/FIDO2 — Information Security capstone

HOTP (RFC 4226) and TOTP (RFC 6238) implemented from the specifications and validated against pyotp as an independent oracle, with an AES-256-GCM encrypted vault, Argon2id hashing, WebAuthn/FIDO2 attestation and assertion verification, and an Isolation Forest layer scoring login risk over geographic distance, impossible travel, new-device and unusual-hour features with LLM-generated alerts.

The detail that carries weight: implemented from the RFCs and validated against an independent oracle, rather than calling a library.

Automated Information Security Risk Assessment Platform

Flask, React, Streamlit, Claude API, NVD API

NIST SP 800-30 risk assessment application: asset, threat and vulnerability registers with live CVE lookups through the NVD API, LLM-generated control recommendations with a rule-based offline fallback, and automated PDF reporting for risk registers, cost-benefit analysis and compliance checklists, served from a Flask REST API backing both Streamlit and React frontends.

COMPLETE SKILLS INVENTORY

The master list. Each CV variant draws a subset; nothing outside this list should appear on any of them.

Programming Languages: Python, C#, TypeScript, JavaScript, SQL, C++, PHP

AI and LLM Engineering: Retrieval-Augmented Generation (RAG), LangChain, LangGraph, agentic AI, tool-calling agents, multi-agent orchestration, prompt engineering, chunking and retrieval tuning, embeddings, hybrid search (BM25 + vector + semantic), RAG evaluation with RAGAS, permission-aware and tenant-scoped retrieval, hallucination mitigation, LangGraph StateGraph design, checkpointing and agent persistence, human-in-the-loop gating, capability-based tool isolation, token streaming over Server-Sent Events, large language models, transformers

Machine Learning: Natural language processing, text classification, sentiment analysis, neural networks, deep learning, anomaly detection (Isolation Forest), feature engineering, model evaluation and deployment, supervised and unsupervised learning, statistical analysis, data visualization

AI Platforms and Libraries: Azure OpenAI Service, Azure AI Search, Azure AI Document Intelligence, Azure AI Foundry, Microsoft Entra ID, Claude API, Ollama / Llama 3.1, scikit-learn, pandas, NumPy

Backend: ASP.NET Core 8, .NET Framework, C#, Entity Framework, FastAPI, Quart, Flask, Node.js, Laravel, REST API design, JWT, RBAC, multi-tenant access control, Server-Sent Events streaming, LiveKit, WebRTC

Frontend: React 18/19, TypeScript, Next.js, Vue.js, Zustand, React Query, Tailwind CSS, Vite, Streamlit, responsive design, component architecture, state management

Databases and Data: Microsoft SQL Server, MySQL, MongoDB, SQLite, relational schema design, schema deployment, stored procedures, query optimization, indexing strategies, ETL, exploratory data analysis, reporting engines

Cloud and DevOps: Microsoft Azure, Azure Blob Storage, Git, GitHub, GitHub Actions, CI/CD

Security Engineering: OAuth/JWT, WebAuthn/FIDO2, TOTP (RFC 6238), HOTP (RFC 4226), AES-256-GCM, Argon2id, NIST SP 800-30, CVE/NVD vulnerability analysis

Practices and Tools: Agile/Scrum, sprint planning, code review, technical documentation, pytest, Postman, Jupyter, Cursor, VS Code

DO NOT LIST — EXCLUSIONS

These are the things that get added back by accident, usually by a well-meaning AI tool filling space. Each is off the CVs deliberately.

- **Microsoft AI-103, in any form.** Removed from all four CVs. Not "in progress", not "studying for", not on a portfolio site. It goes back only when it is passed.
- **Any percentage or performance metric.** No accuracy improvements, no latency reductions, no "improved X by Y%", no cost savings, no time savings. Nothing was benchmarked, so nothing can be claimed.
- **User counts, request volumes, uptime figures, data volumes.** Unknown. Do not estimate them.
- **RAGAS scores.** The harness is built; it has not been run. Describe the instrumentation, never the results, until the run happens.
- **Project-Ease as complete or shipped.** It is in development, and every mention of it must say so.
- **Team-lead, management or mentoring language.** The role is individual contributor. "Sole engineer on a module" is the accurate and stronger claim; "led a team" is neither.
- **Inflated seniority.** The title is Associate Software Engineer. Do not soften it to "Software Engineer" or stretch it to "Senior" anywhere, including LinkedIn.
- **Padding technologies.** Nothing goes on the skills list that has not actually been used in the work described above. Docker, Kubernetes, AWS, GCP, Spark, Kafka and similar are absent for that reason.
- **Coursework dressed up as projects.** The two capstones are listed as what they are. Do not invent additional academic projects to fill space.

OPEN GUARDRAILS

Settle these before this document is pointed at a portfolio site, a new CV variant or an interview. They are open questions, not errors.

1. The LangGraph scope question — RESOLVED, August 2026.

Confirmed: the LangChain and LangGraph agentic pipelines genuinely run inside the Nucleus One production platform. The opening bullet of the AI/ML CV stands as written, and the same claim is safe to use on a portfolio site, on LinkedIn and in interviews. No rewrite needed.

Because it is the first line an interviewer reads, it will also be the first thing they open with. Expect "walk me through the LangGraph pipelines in your production platform" and be ready with specifics: which workflows the agents serve, which custom tools were defined and why, where the human-in-the-loop approval step sits and what it gates, how state is carried across steps, and how the pipelines talk to the ASP.NET Core 8 APIs and the SQL Server layer. The claim is true — the interview turns on how concretely it can be described.

2. The RAGAS run — highest-value remaining item.

One evening on Project-Ease produces real faithfulness, answer relevancy and context precision numbers. That validates a claim already sitting on the CV and the README, converts "wired in an evaluation harness" into "measured and improved retrieval quality against it", and would be the only measured result in the entire portfolio. Highest return on time of anything left.

3. Consistency across surfaces.

LinkedIn, the GitHub profile, the portfolio site and these four CVs should agree on dates, title and the AI-103 removal. A mismatch between a CV and a public profile is noticed and it is expensive. Check all four surfaces after any change here.

4. README quality on the two flagship repositories.

RepoWarden and Project-Ease are linked from every CV variant, so the README is part of the application. Architecture diagram, the capability-isolation argument stated plainly, setup instructions that work from a clean clone. An interviewer who clicks through and finds a default README loses the strongest evidence in the pack.